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MyDesign Portfolio

Element E Initial Template

Element E: Application of STEM Principles and Practices

STEM Principles and Practices to Design Requirements (E1)

STEM Principle for High-Priority Design Requirement #1

STEM Concept: **Science- Physics**

Physics uses forces like balance and gravity to predict movement and outcomes, which can be applied to our problem.

STEM Principle for High-Priority Design Requirement #2

STEM Concept: **Math - Measurements and Dimensions**

Any design that is created comes with its dimensions and measurements in height, width, and depth.

STEM Principle for High-Priority Design Requirement #3

STEM Concept: **Engineering - Design**

Engineering is everywhere! Anytime someone brainstorms to conclude for a solution to a given predicament, they are engineering!

Proposed Solution to STEM Principles and Practices (E2)

STEM Principle for Design Solution #1

STEM Concept: **Math- Measurements and Dimensions**

Our design had to be measured to be able to have the appropriate height, width, and depth to be able to reach up to get and knock off the birdies.

STEM Principle for Design Solution #2

STEM Concept: **Science- Physics**

Describe how the STEM concept applies directly to the specifics of your solution: **Our long stick has to have just the right amount of weight at the top to be able to be strong**

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enough to knock the birdies down, but light enough for it not to bend over to the force of gravity applying on it.

STEM Principle for Design Solution #3

STEM Concept: **Engineering- Design**

Describe how STEM concept applies directly to the specifics of your solution:

We need to design an object to remove the birdies from the gym roof and windows. We diminished the amount of ideas we had, and narrowed it down to a long cardboard stick. We concluded that it would be the most effective and most resourceful.

Evidence Design is Reviewed (E3)

*Documentation #1: **Mr. Steinhorn***

- What format: **Face to face conversation**
- What the expert said about the connection you've made from your design requirements to STEM principles and practices: Mr. Steinhorn explained to us that we must focus on design principles during our construction. His experience in the engineering and technology field gave us insight on tips for our building, like how we should have multiple parts and tape them together for construction.
- What the expert said about how STEM principles and practices substantiate your design solution: Mr. Stienhorn wanted us to use

*Documentation #2: **Mrs. Armstrong***

- What format: **Facce to face conversation**
- What the expert said about the connection you've made from your design requirements to STEM principles and practices: **Physics are a crucial part to experimenting how to develop an efficient collapsing system.**
- What the expert said about how STEM principles and practices substantiate your design solution: **We can use physics to develop a well rounded design- in this case our cardboard stick. Mrs. Armstrong is a mechanical engineer who works with design principles and physics every day.**

*Documentation #3: **Mr. Butchko***

- What format: Interview notes, emails, audio/video tape with transcript, etc.



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Face to face conversation

- What the expert said about the connection you've made from your design requirements to STEM principles and practices: **The expert said that we need multiple supports for the card board as it is not stable. He also recommended that we should have a strong base.**
- What the expert said about how STEM principles and practices substantiate your design solution: **Mr. Butchko is a civil engineer and uses engineering principles every day to design buildings. He said that we should use math and measurements with engineering and construction to create the best design.**

Experts with Credentials (E4)

Document who your field experts are (names are not necessary) and what qualifies them as an expert. Use as many copies of this sub-template as needed.

Expert #1: Handled multiple engineering projects in electronics and IT.

Expert #2: Is a mechanical engineer who works with design principles.

Expert #3: A civil engineer who is an expert in consulting and advising construction.

Verification or Corrective Response to Design Review (E5)

Your conversation/interview with the expert(s) should inform some corrective responses. In E5, describe how those reviews informed your responses. For example, "_____ told us _____, which informed our decision to change/reconsider _____." (or "...which verified/reinforced our decision to _____."

Make sure to document the experts' reviews (suggestions/feedback or criticisms of your proposed solution) and how they will inform iterations you will make to your proposed solution. Use as many copies of this sub-template as needed.

Interview with Expert #1 informs corrective response:

Mr. Steinhorn asked us to have a solid design before we started to build, which changed our way of thinking when it came to starting construction and to value patience over speed.

Interview with Expert #2 informs corrective response:



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Mrs. Armstrong told us to think about the requirements of the project and the height of the birdies. This lead us to double check our measurements and reassured us of our designs effectiveness.

Interview with Expert #3 informs corrective response:

Mr. Butchko said that our structure wasn't stable enough. This lead us to change our base design and add reinforcements to our design.